HWA CHONG INSTITUTION
JC2 Preliminary Examinations
Higher 2

CANDIDATE NAME		**CT GROUP**	13S_____
CENTRE NUMBER		**INDEX NUMBER**	

BIOLOGY — 9648 / 03

Paper 3 Applications Paper and Planning Question — **17 September 2014**

Additional Material: Writing Paper — **2 hours**

INSTRUCTIONS TO CANDIDATES

There are **four** question booklets (**I** to **IV**) to this paper. Write your **name**, **CT group**, **centre number** and **index number** in the spaces provided at the top of this cover page, and your **name** and **CT group** on the lines provided at the top of the cover page of Booklets **II, III and IV**.

STRUCTURED QUESTIONS

Answer **all** three questions. Write your answers on the lines provided.

PLANNING QUESTION

Answer the question in booklet **IV**. Write your answers on the lines / in the spaces provided.

FREE RESPONSE QUESTION

Answer the question. Your answers must be in continuous prose, where appropriate.

Write your answers on the writing paper provided.

BEGIN EACH PART ON A FRESH SHEET OF WRITING PAPER.

A **NIL RETURN** is required.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.

Calculators may be used.

You are reminded of the need for good English and clear presentation in your answers.

For Examiners' Use	
Question	**Marks**
1	**/ 13**
2	**/ 13**
3	**/ 14**
4	**/ 12**
5	**/ 20**
Total	**/ 72**

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BOOKLET I

STRUCTURED QUESTIONS

QUESTION 1

Variable number tandem repeats (VNTRs) are useful for various applications including DNA fingerprinting and paternity testing. Genetic polymorphisms at VNTR loci are detected by restriction fragment length polymorphism (RFLP) analysis.

(a) State what is meant by *VNTRs*. [2]

VNTR loci analysis reveals genetic relationships within families. In a particular family with three children, one child was diagnosed with an autosomal dominant disease, although both parents were normal. To determine if the father was the biological father, two VNTR loci *Blu* and *Gry* were analysed.

Restriction enzymes were used to yield four alleles at the *Blu* locus, **B1**, **B2**, **B3** and **B4**, and two alleles at the *Gry* locus, **G1** and **G2** as shown in Fig. 1.1. The repeat units of the two loci contain the same number of nucleotides.

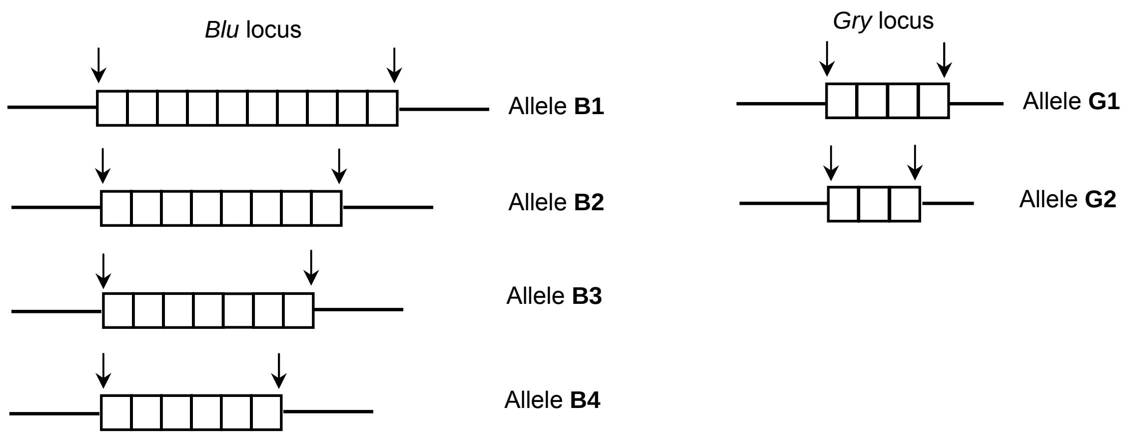

Fig. 1.1

Fig. 1.2 shows the resulting pattern of bands on a Southern blot. The genotype of the mother is **B1B4G2G2**.#

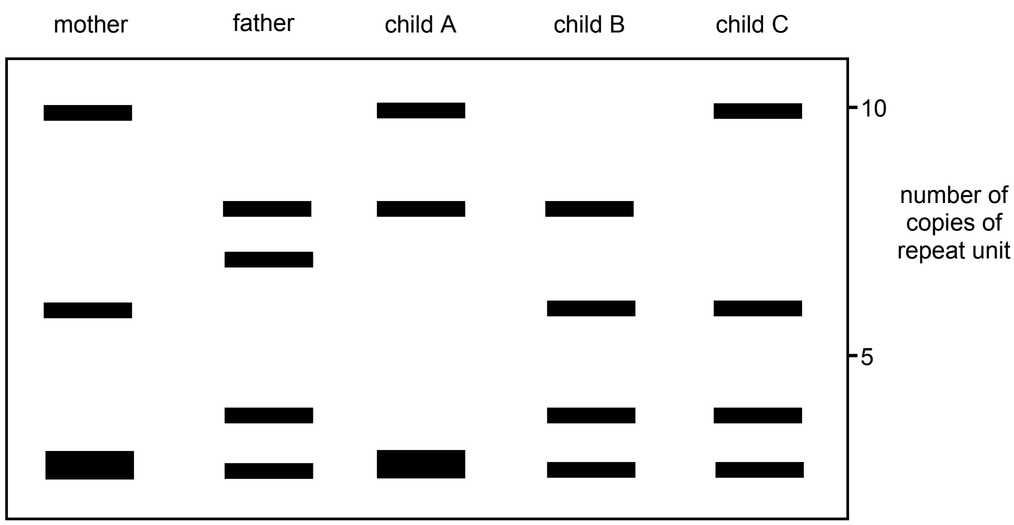

Fig. 1.2

(b) With reference to Fig. 1.1 and Fig. 1.2,

(i) explain how gel electrophoresis is used to separate the different VNTR alleles. [4]

(ii) identify the child who may not be an offspring of the father in this family. [1]

(iii) account for your answer in **(b)(ii)**. [2]

Several hundred VNTR loci have been discovered and mapped in the human genome. With the completion of the Human Genome Project (HGP), more than 1,800 disease genes have been discovered. A survey revealed that 15% of respondents have undergone genetic testing in a medical setting.

(c) Discuss how the HGP may be beneficial to individuals undergoing genetic testing in a medical setting. [2]

(d) Suggest two ways by which information gained from genetic testing may be misused. [2]

[Total: 13]

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BOOKLET II

STRUCTURED QUESTIONS (continued)

QUESTION 2

In 2014, a team of scientists claimed that they were able to reprogramme somatic cells into pluripotent stem cells through exposure to stimuli. These resultant stem cells were termed "stimulus-triggered acquisition of pluripotency (STAP) stem cells".

Fig. 2.1 shows how a study was conducted to test if STAP stem cells (STAP SCs) are pluripotent. Somatic cells were isolated from mice and exposed to an acidic environment to form STAP SCs. The STAP SCs were then injected into an early-stage mouse embryo.

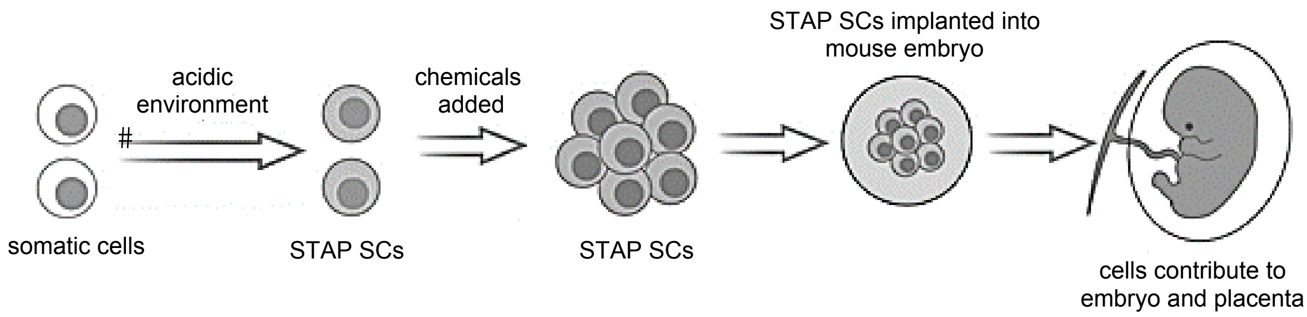

Fig. 2.1

(a) Suggest the role of the acidic environment in the formation of STAP SCs. [1]

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(b) With reference to Fig. 2.1 and the information provided above, describe the unique features of STAP SCs. [2]

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(c) Explain whether this study shows that the STAP SCs are pluripotent. [2]

Stem cells are often used in gene therapy to treat severe combined immunodeficiency (SCID).

(d) Describe the genetic basis of SCID. [3]

In a study, two methods of gene transfer, *ex vivo* and *in vivo*, were employed to treat children suffering from SCID. The results of the study were shown in Table 2.1.

Table 2.1

method of gene transfer	number of cells expressing the normal allele	
	before treatment	after treatment
ex vivo	1.0×10^6	8.0×10^6
in vivo	1.0×10^6	2.5×10^6

(e) Describe which method of gene transfer is more effective in delivering the transgene to the children suffering from SCID. [3]

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(f) Suggest two possible reasons for the difference in the effectiveness of these methods of gene transfer. [2]

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[Total: 13]

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BOOKLET III

SECTION A: STRUCTURED QUESTIONS (continued)

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QUESTION 3

Agrobacterium is a soil bacterium used in genetic engineering of plants. Fig. 3.1 shows the stages in the transfer of a *Bacillus thuringiensis* (*Bt*) gene into plant cells.

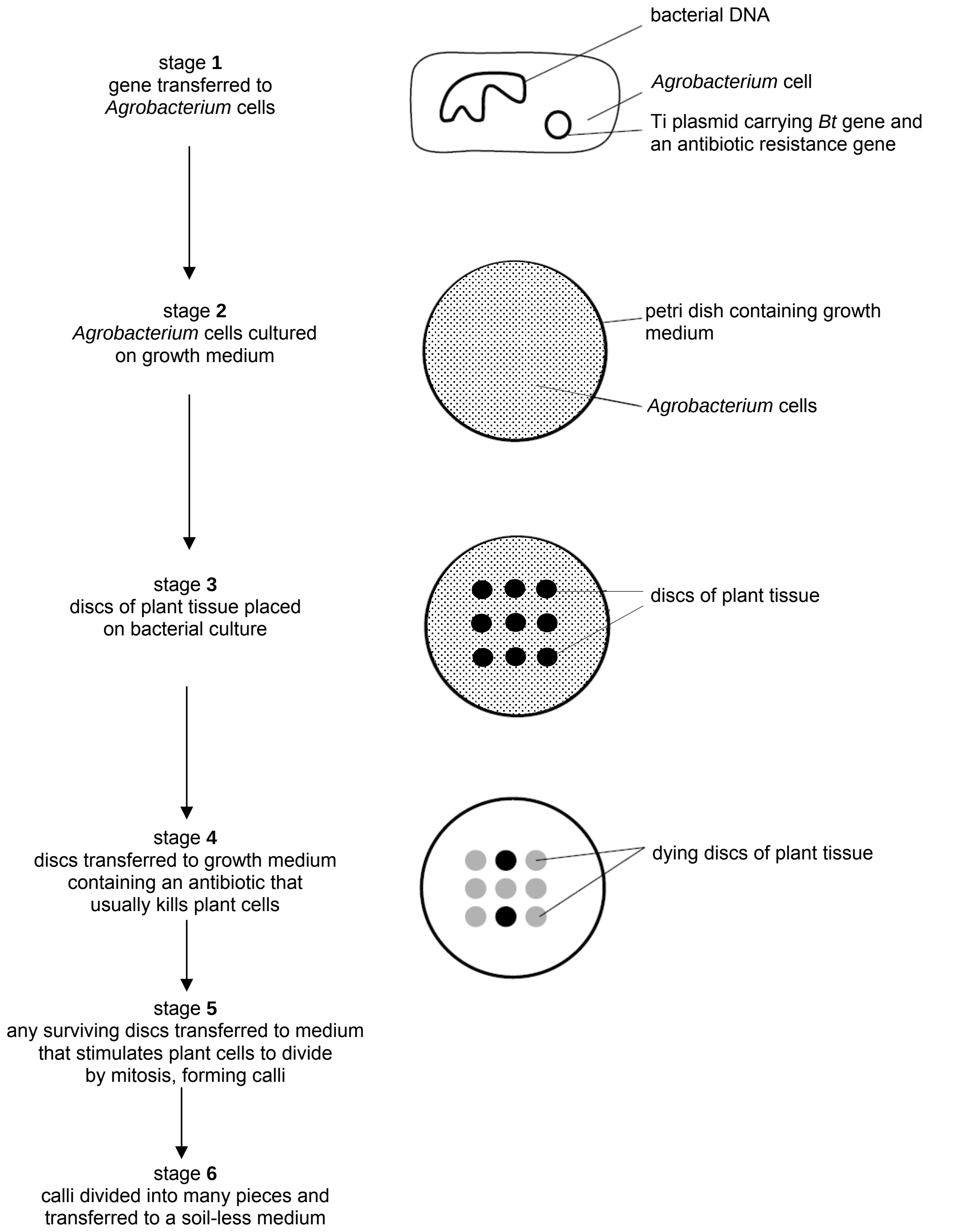

Fig. 3.1

(a) Explain how the Ti plasmid carrying the *Bt* gene is transferred into the *Agrobacterium* cells. [2]

(b) With reference to Fig. 3.1, explain the significance of each of the two stages.

stage **2**: [2]

stage **6**: [2]

(c) Suggest why the genetic engineering of crop plants as shown in Fig. 3.1 is more favourable than spraying crops with pesticides. [1]

YieldGard, a type of Bt corn that produces a single type of Bt protein, was planted in Midsouth, USA. A Bt resistance monitoring programme showed that the frequency of resistance to the Bt protein in YieldGard increased significantly in sugarcane borers (SCBs). SCBs not only feed on sugarcane, but also other crops such as corn.

To counter the Bt-resistant SCBs, Bt corn expressing multiple types of Bt proteins were recently employed in a greenhouse trial. The trial included a non-Bt control corn and three types of Bt corn:

- Non-Bt control corn – does not produce Bt protein,
- YieldGard®– produces a single type of Bt protein,
- Genuity® VT Triple Pro™ – produces three types of Bt proteins, and
- Genuity® SmartStax™ – produces six types of Bt proteins.

Newly hatched larvae of three SCB populations were manually placed on mature corn plants in the greenhouse. The larvae often bore into the corn stalk and develop tunnels within them. Larval tunnelling by borers inside each stalk was recorded after 21 days as shown in Fig. 3.2.

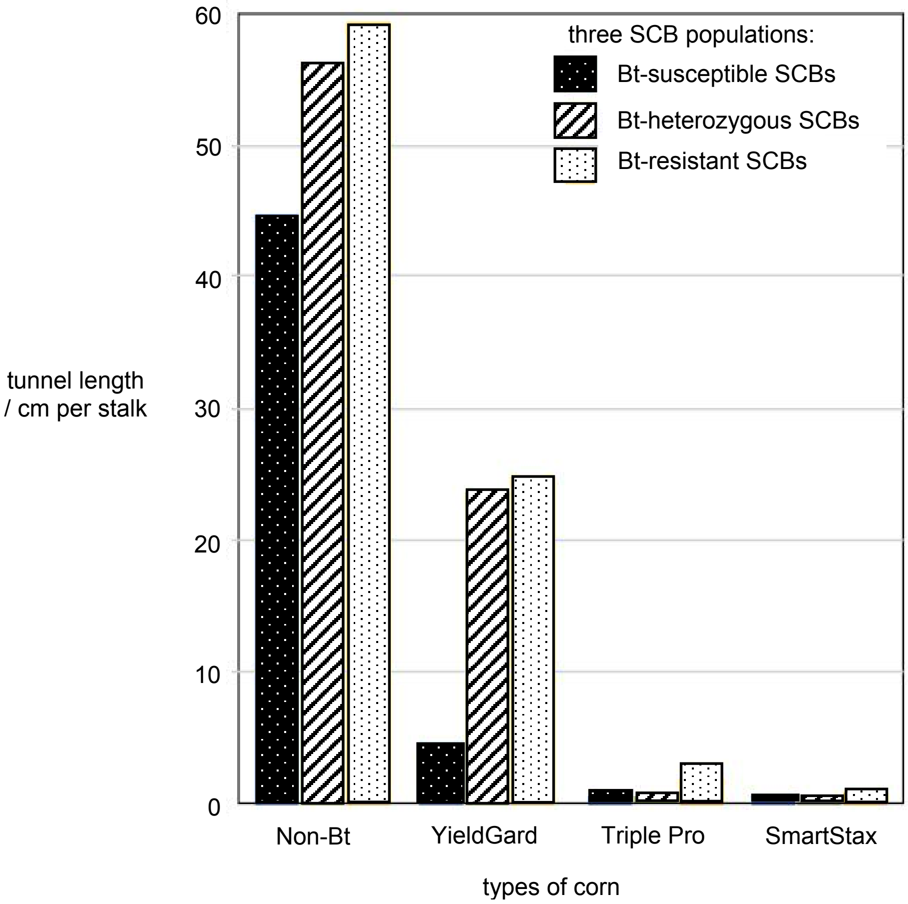

Fig. 3.2

(d) Suggest and explain which type of Bt corn will best increase crop yield. [2]

(e) Explain why YieldGard is no longer optimal for cultivation. [2]

(f) Discuss the need to compare among the three different populations of SCBs. [2]

(g) Suggest how VT Triple Pro and SmartStax are useful in managing Bt resistance in SCBs. [1]

[Total: 14]

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BOOKLET IV

PLANNING QUESTION

QUESTION 4

The enzyme amyloglucosidase hydrolyses the α(1,4) and α(1,6) glycosidic bonds in starch to release glucose.

Copper sulfate binds to a site on amyloglucosidase other than the active site and changes the conformation of the active site.

1% amyloglucosidase is immobilised in sodium alginate beads forming amyloglucosidase-alginate (AGA) beads. A fixed number of AGA beads of uniform shape and size are then packed in a column as shown in Fig. 4.1.

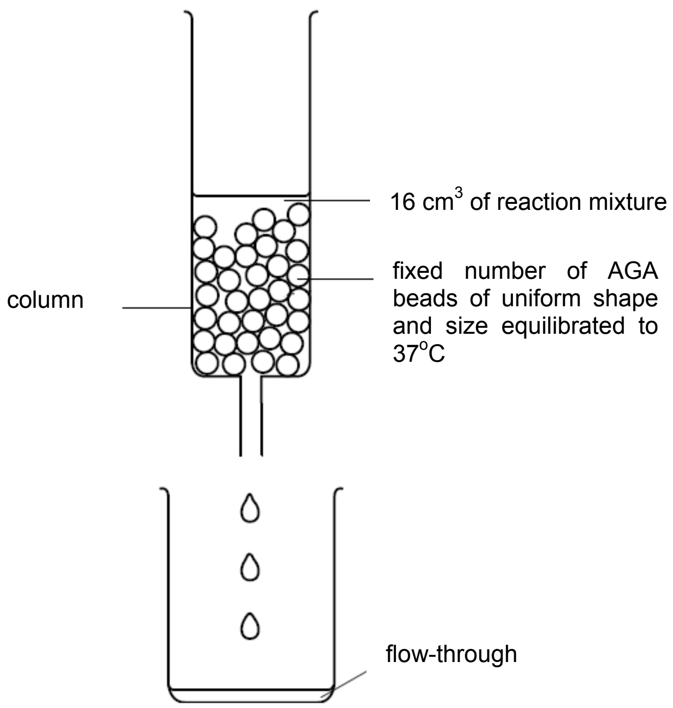

Fig. 4.1

In each run of the experiment, 16 cm^3 of reaction mixture is poured through the column. Glucose solution that is immediately collected is known as flow-through. It takes five minutes for all the flow-through to be collected. Glucose is a reducing sugar that gives an orange-red precipitate with Benedict's solution.

Using this information and your own knowledge, design an experiment to investigate the effect of increasing the concentration of copper sulfate on the rate of hydrolysis of starch by amyloglucosidase.

You must use:

- 0.3% copper sulfate solution,
- 5% starch,
- distilled water,
- Benedict's solution,
- pH 7.0 buffer,
- apparatus shown in Fig. 4.1,
- drying oven / desiccator,
- weighing balance.

You should select from the following apparatus and use appropriate additional apparatus:

- normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, glass rods etc.,
- syringes,
- filter paper and filter funnel,
- Bunsen burner with tripod, gauze and bench mat,
- timer e.g. stopwatch or stop clock,
- thermometer,
- retort stand with clamps.

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- show how you will record your results and the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 12]

FREE RESPONSE QUESTION

Your answers must be in continuous prose, where appropriate.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Write your answers in the writing paper provided.
BEGIN EACH PART ON A FRESH SHEET OF WRITING PAPER.
A **NIL RETURN** is required.

QUESTION 5

(a) Explain the role of restriction enzymes in the formation of a recombinant DNA molecule. [6]#

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(b) Explain why the maintenance of gene libraries is important. [4]#

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(c) Describe how dwarfism in humans can be cured by recombinant DNA technology, and outline the advantages of this cure. [10]#

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[Total: 20]

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